

GENERAL CHEMISTRY



Chapter 11 Liquids and Intermolecular Forces

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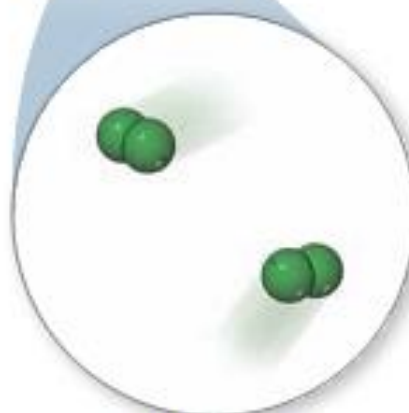
10-1 A Molecular Comparison of Gases, Liquids, and Solids

Gas	Liquid	Solid
Assumes the shape and volume of its container	Assumes the shape of the portion of its container it occupies, but not the volume	Maintains a fixed shape and volume
Highly compressible	Not really compressible	Not compressible
Viscosity Low viscous	Viscosity Moderate viscous	Viscosity High viscous

Strength of intermolecular attractions increasing



Gas

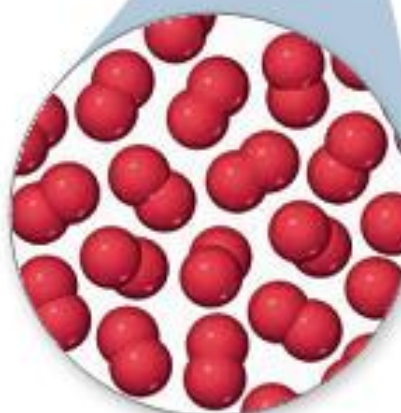


Chlorine, Cl_2

Particles far apart; possess complete freedom of motion



Liquid

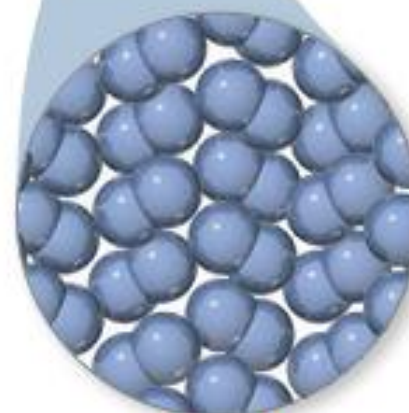


Bromine, Br_2

Particles are closely packed but randomly oriented; retain freedom of motion; rapidly change neighbors



Crystalline solid



Iodine, I_2

Particles are closely packed in an ordered array; positions are essentially fixed

11-2 Intermolecular Forces

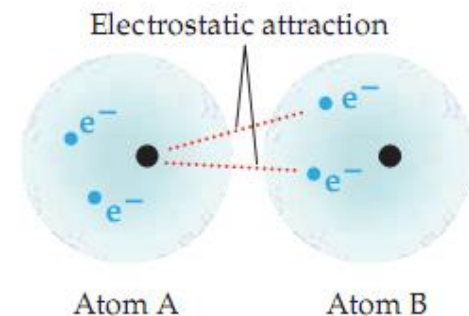
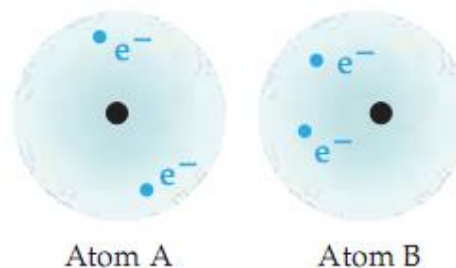
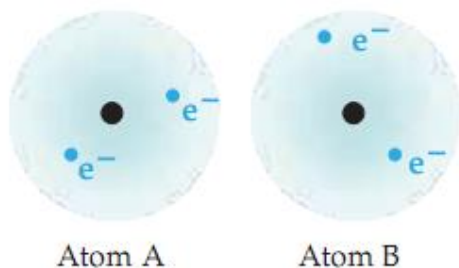
TABLE 11.2 Melting and Boiling Points of Representative Substances

Force Holding Particles Together	Substance	Melting Point (K)	Boiling Point (K)
<i>Chemical bonds</i>			
Ionic bonds	Lithium fluoride (LiF)	1118	1949
Metallic bonds	Beryllium (Be)	1560	2742
Covalent bonds	Diamond (C)	3800	4300
<i>Intermolecular forces</i>			
Dispersion forces	Nitrogen (N ₂)	63	77
Dipole-dipole interactions	Hydrogen chloride (HCl)	158	188
Hydrogen bonding	Hydrogen fluoride (HF)	190	293

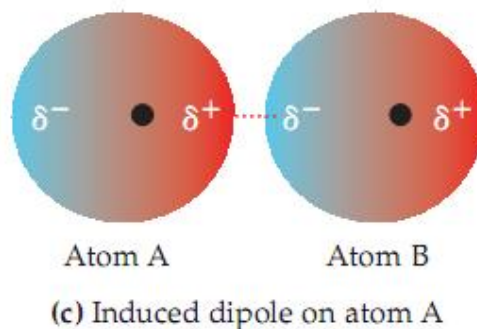
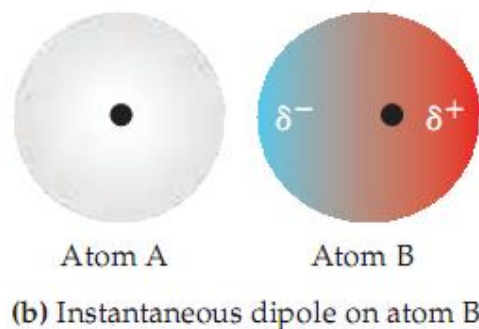
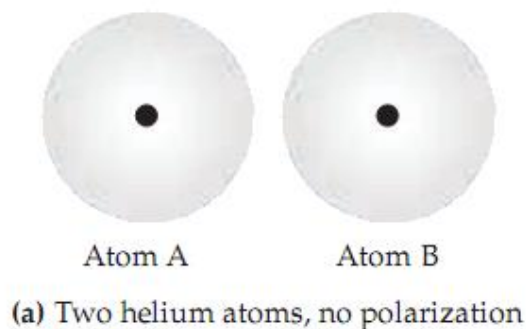
van der Waals forces

Dispersion Forces

Subatomic view

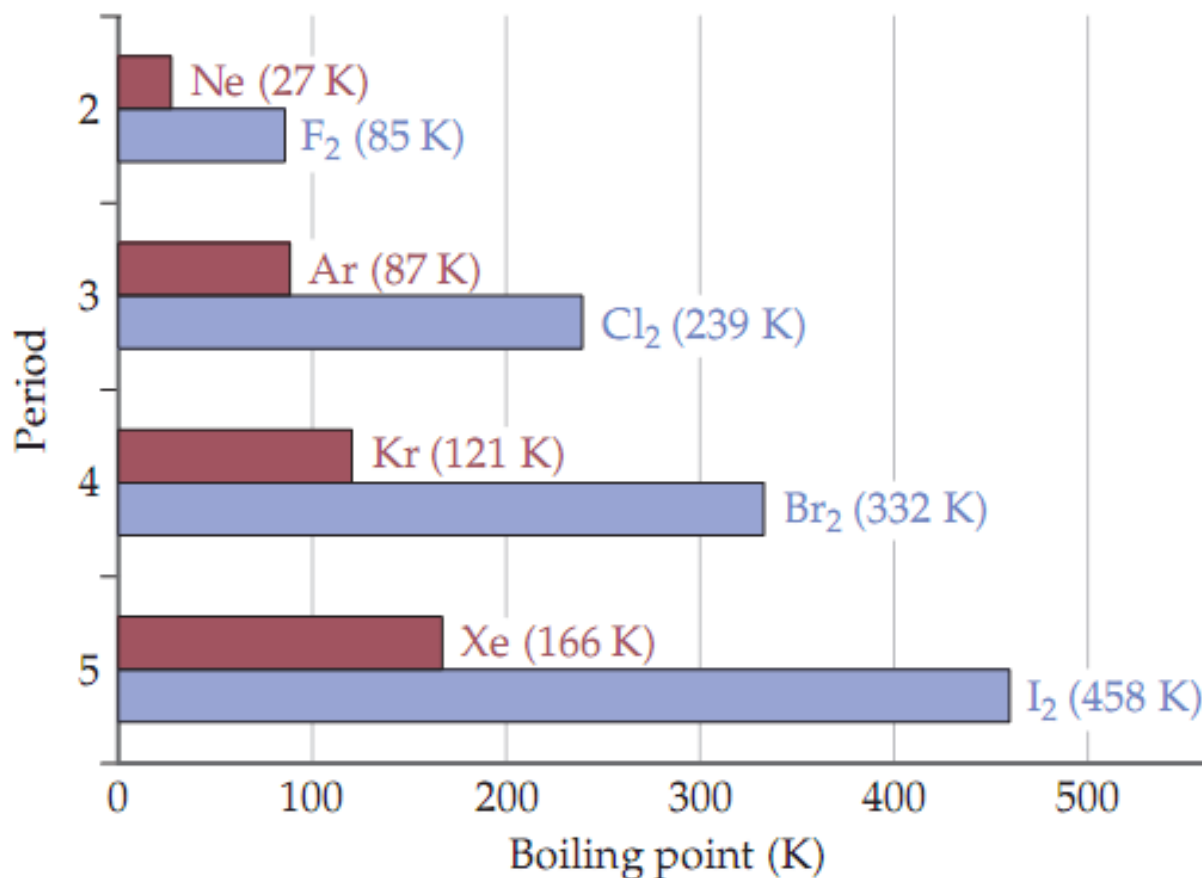


Polarization view



London dispersion
forces or induced
dipole–induced dipole
interactions

Dispersion forces tend to increase in strength with increasing molecular weight.



Molecular shape affects intermolecular attraction

Linear molecule—larger surface area enhances intermolecular contact and increases dispersion force



n-Pentane (C₅H₁₂)
bp = 309.4 K

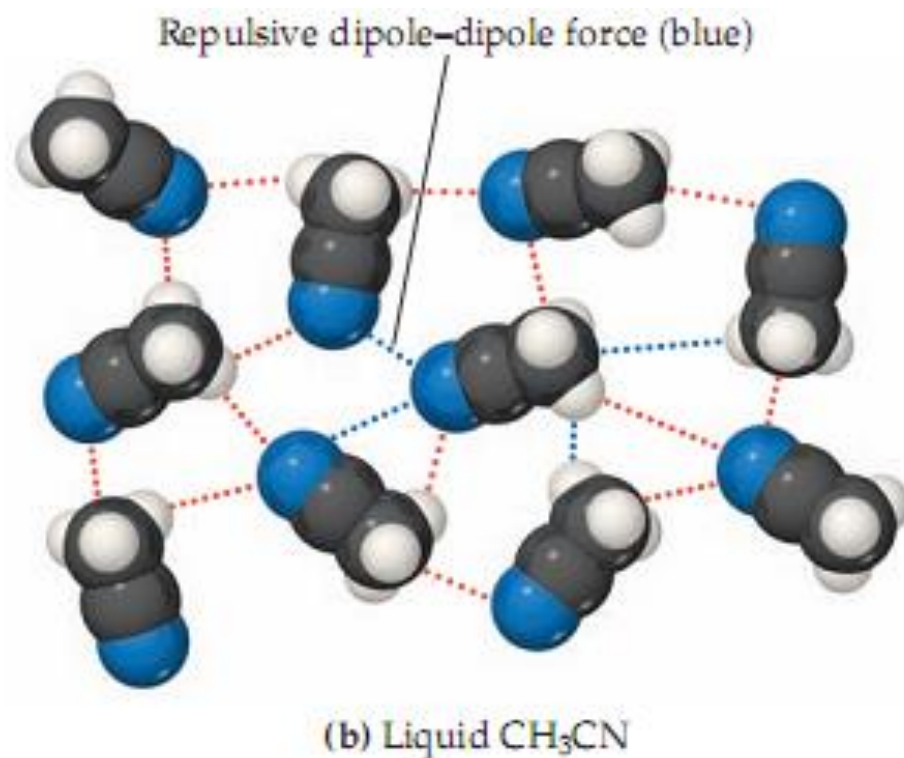
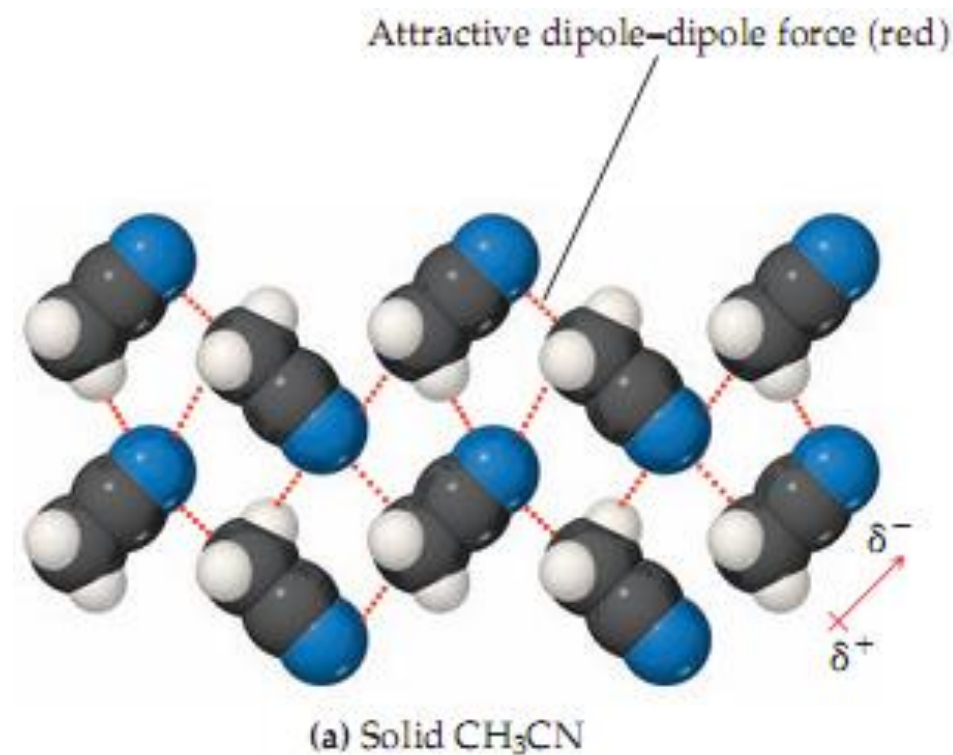
Spherical molecule—smaller surface area diminishes intermolecular contact and decreases dispersion force



Neopentane (C₅H₁₂)
bp = 282.7 K

Dipole–Dipole Interactions

- ◆ dipole–dipole interactions originate from electrostatic attractions between the partially positive end of one molecule and the partially negative end of a neighboring molecule.



Moving from left to right, do the dispersion forces get stronger, get weaker, or stay roughly the same in the molecules shown here?



Propane
CH3CH2CH3
MW = 44 amu
 $\mu = 0.1$ D
bp = 231 K



Dimethyl ether
CH3OCH3
MW = 46 amu
 $\mu = 1.3$ D
bp = 248 K



Acetaldehyde
CH3CHO
MW = 44 amu
 $\mu = 2.7$ D
bp = 294 K



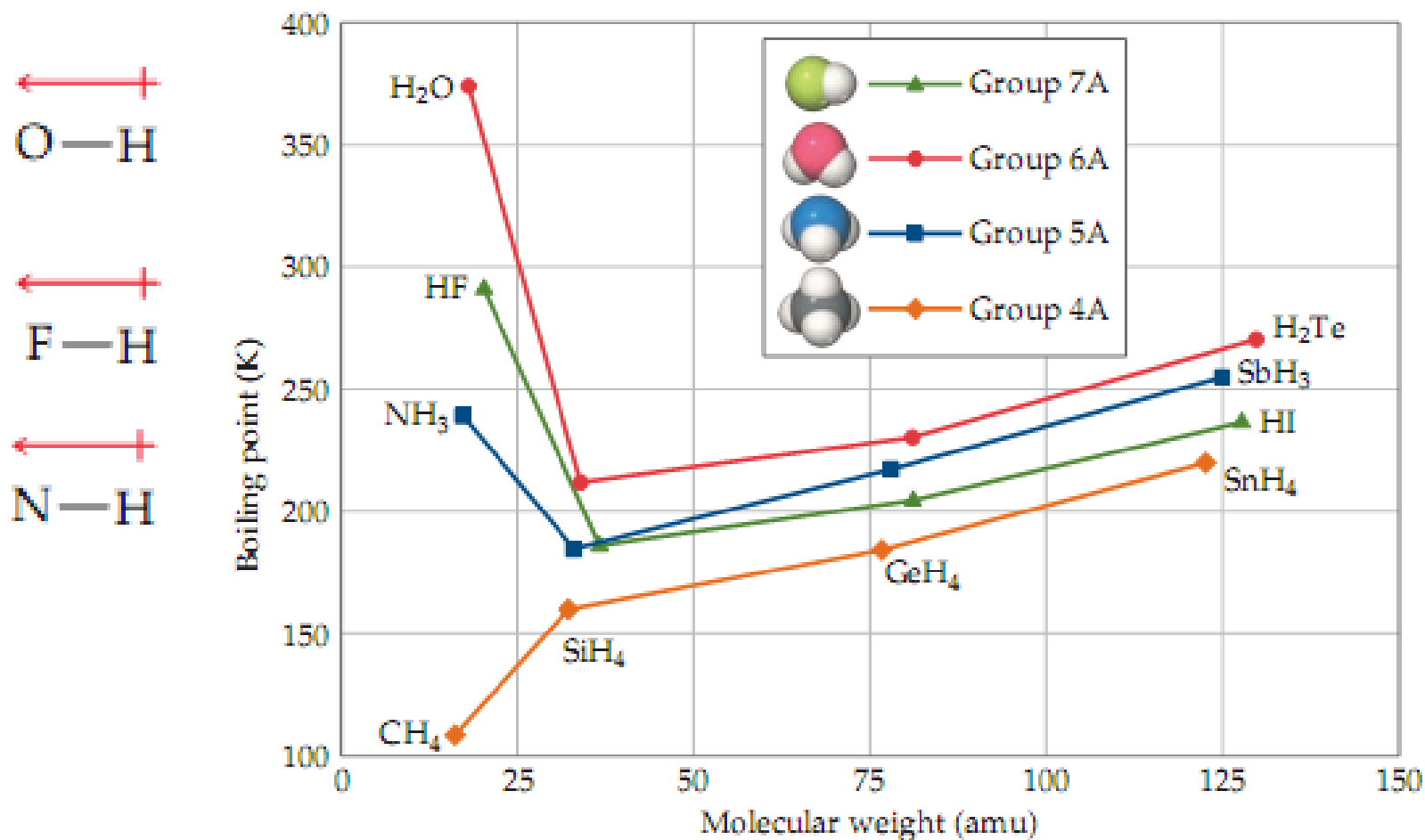
Acetonitrile
CH3CN
MW = 41 amu
 $\mu = 3.9$ D
bp = 355 K

For molecules of approximately equal mass and size, the strength of intermolecular attractions increases with increasing polarity

Increasing polarity
Increasing strength of dipole-dipole forces

Hydrogen Bonding

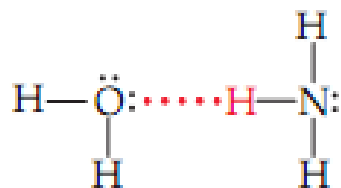
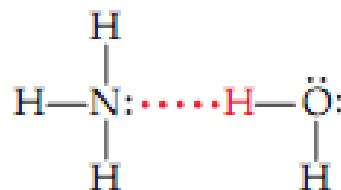
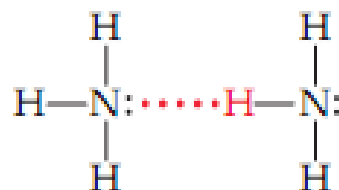
A hydrogen bond is an attraction between a hydrogen atom attached to a highly electronegative atom (usually F, O, or N) and a nearby small electronegative atom in another molecule or chemical group



Covalent Bonds vs Hydrogen Bonds

Covalent bond,
intramolecular

Hydrogen bond,
intermolecular



EXAMPLE

In which of these substances is hydrogen bonding likely to play an important role in determining physical properties: methane (~~CH₄~~), hydrazine (H₂NNH₂), methyl fluoride (~~CH₃F~~), hydrogen sulfide (~~H₂S~~)

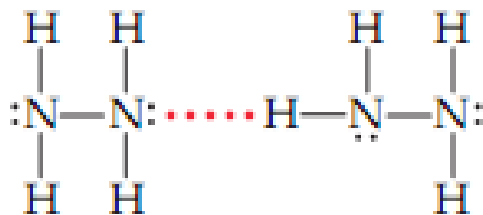
CH₄ and H₂S do not contain H bonded to N, O, or F

➔ Eliminate CH₄ and H₂S

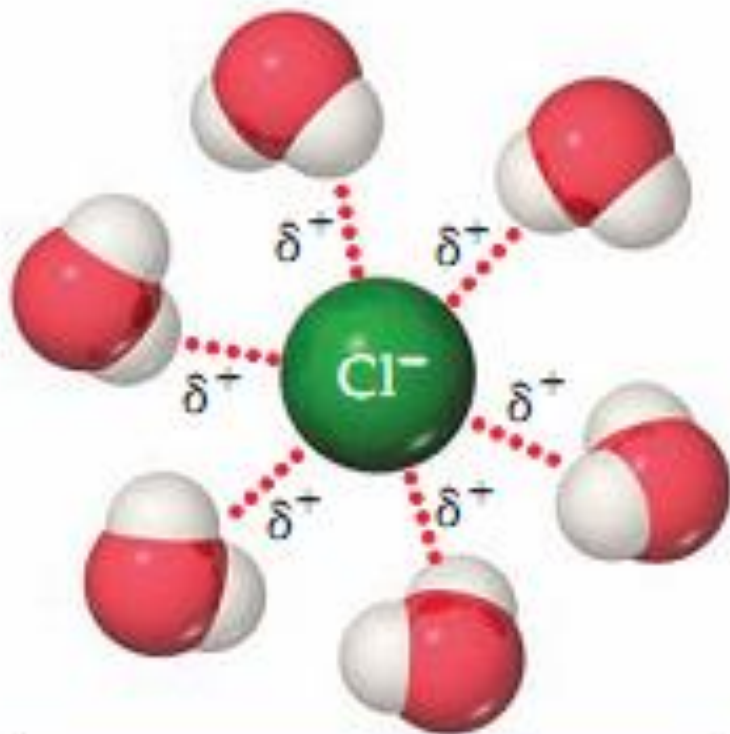
CH₃F contain C – H and C – F bonds, not H – F bond

➔ Eliminate CH₃F

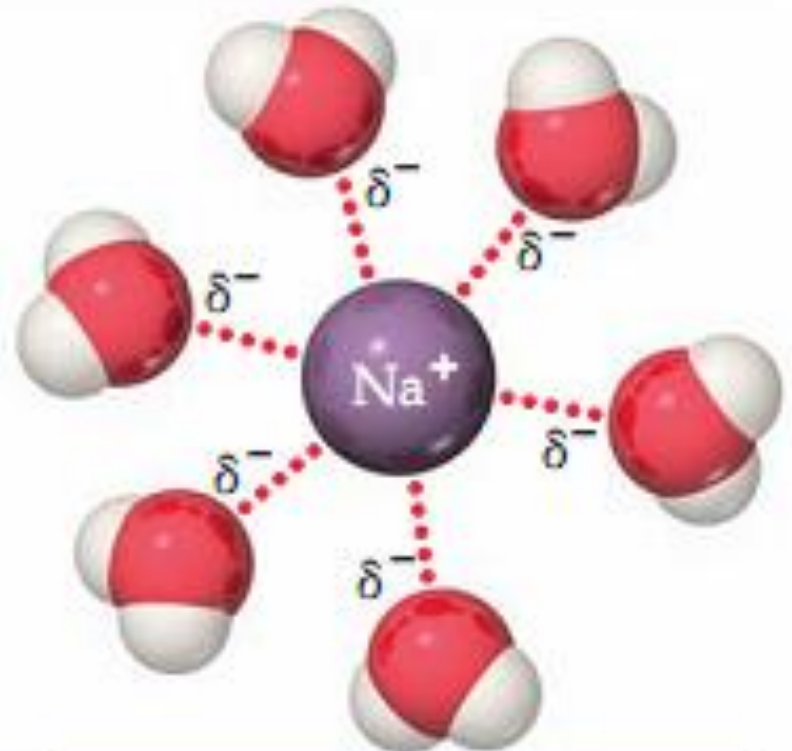
H₂NNH₂ contain N – H bond, and each N has a lone pair



Ion–Dipole Forces



Positive ends of polar molecules are oriented toward negatively charged anion

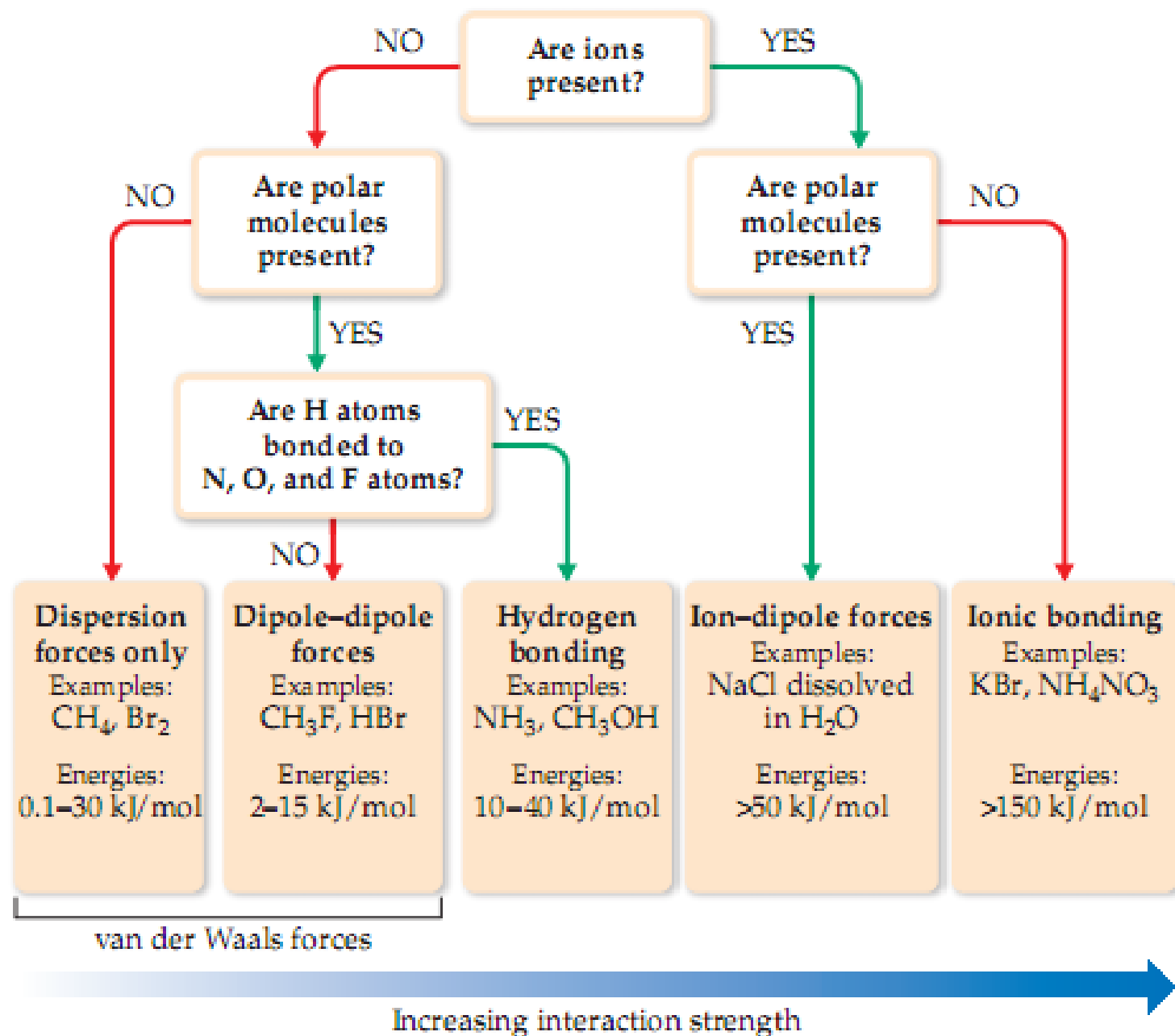


Negative ends of polar molecules are oriented toward positively charged cation

Comparing Intermolecular Forces

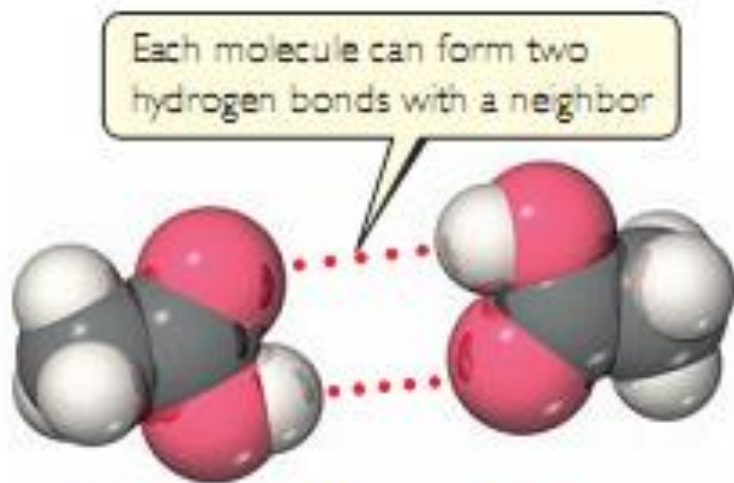
1. When the molecules of two substances have comparable molecular weights and shapes, dispersion forces are approximately equal in the two substances.
2. When the molecules of two substances differ widely in molecular weights, and there is no hydrogen bonding, dispersion forces tend to determine which substance has the stronger intermolecular attractions.

Flowchart for determining intermolecular forces

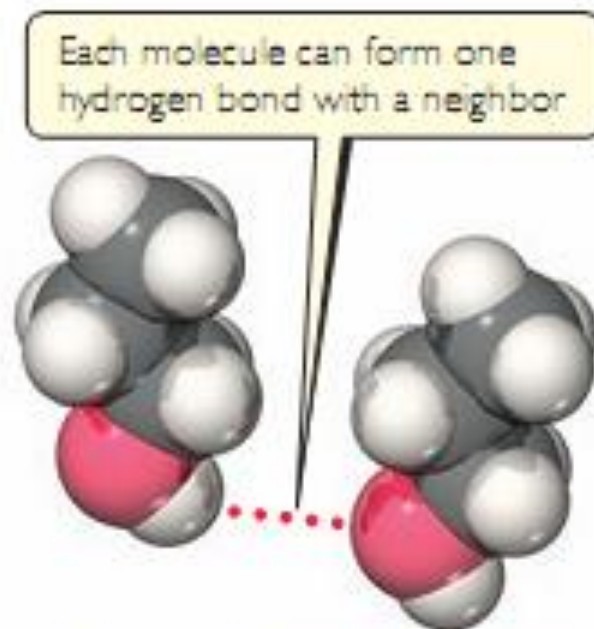


Comparing Intermolecular Forces

The greater the number of hydrogen bonds possible, the more tightly the molecules are held together and, therefore, the higher the boiling point



Acetic acid, CH_3COOH
MW = 60 amu
bp = 391 K



1-Propanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
MW = 60 amu
bp = 370 K

EXAMPLE

List the substances BaCl_2 , H_2 , CO , HF , and Ne in order of increasing boiling point.



- The attractive forces are stronger for ionic substances than for molecular ones → BaCl_2 should have the highest boiling point
- The intermolecular forces of the remaining substances depend on
- molecular weight, H_2 : 2; CO : 28; HF : 20; and Ne : 20
- polarity, H_2 , Ne : non polar ; CO : slightly polar, HF : polar
- hydrogen bonding HF can hydrogen bond: polar

11-3 Select Properties of Liquids

Viscosity



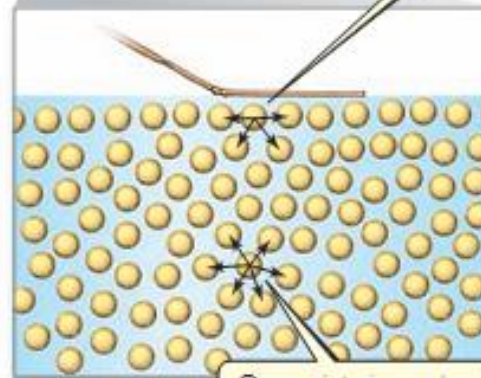
SAE 40
higher number
higher viscosity
slower pouring

SAE 10
lower number
lower viscosity
faster pouring

Surface Tension

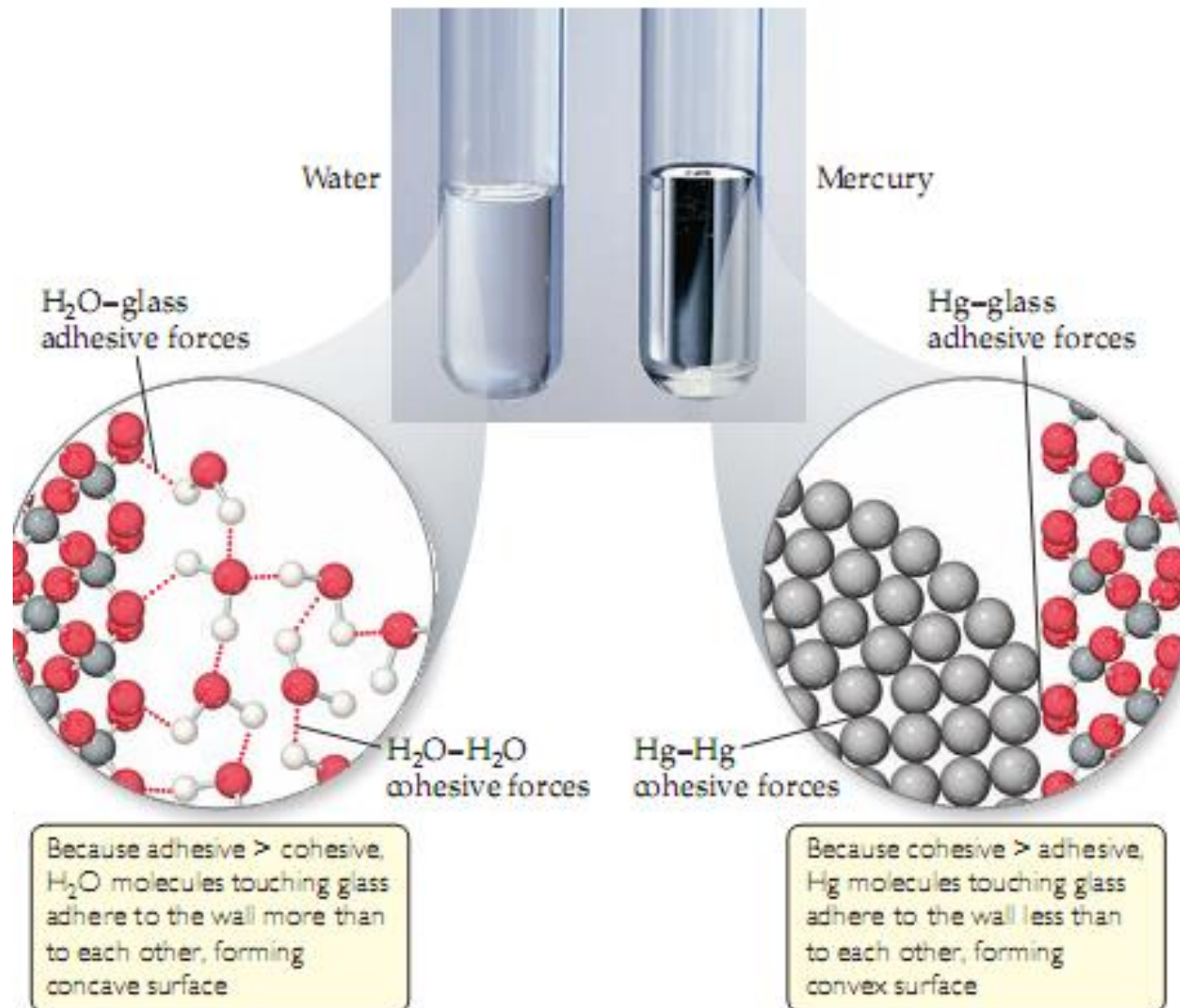


On any surface molecule, there is no upward force to cancel the downward force, which means each surface molecule "feels" a net downward pull

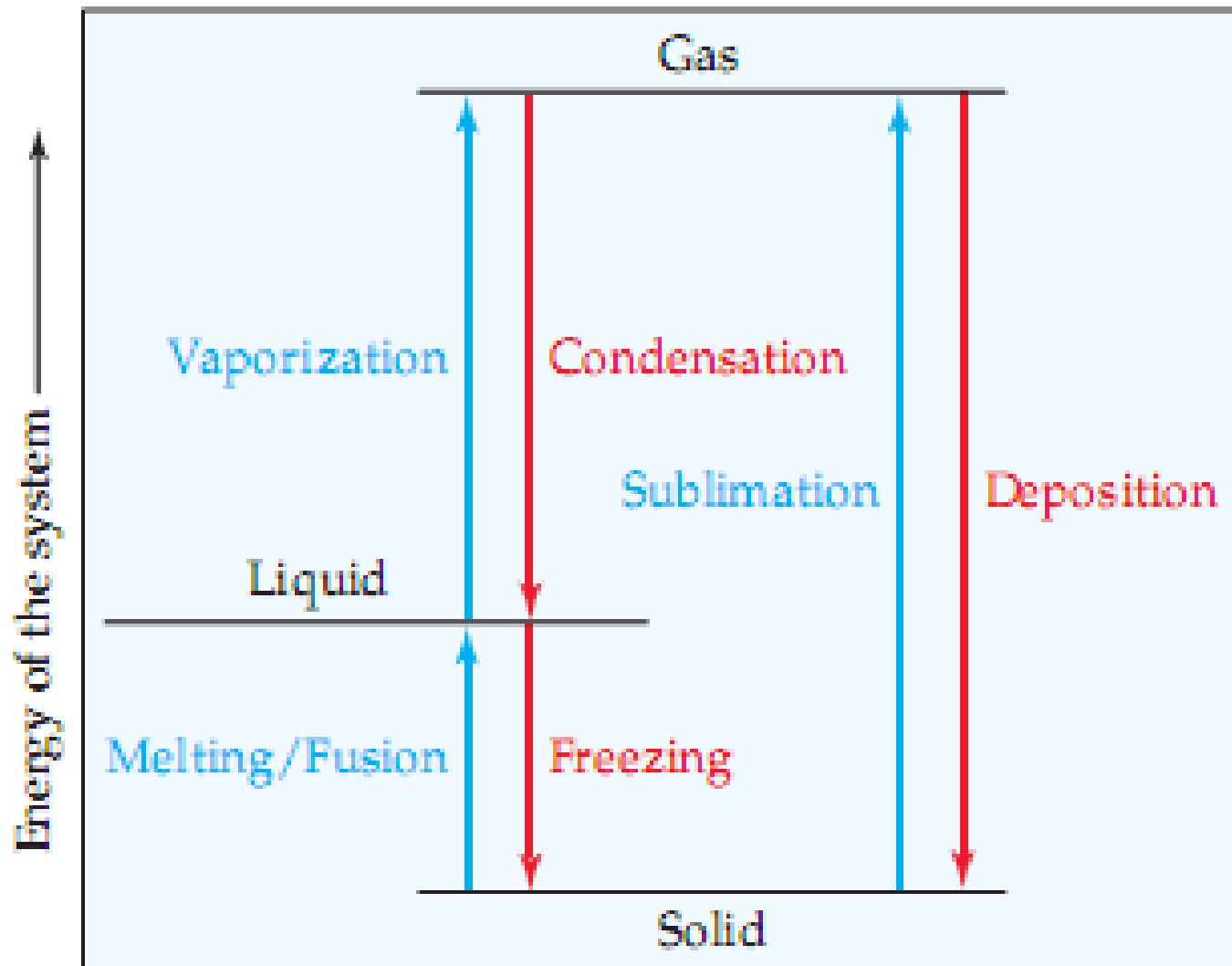


On any interior molecule, each force is balanced by a force pulling in the opposite direction, which means that interior molecules "feel" no net pull in any direction

Capillary Action



11-4 Phase Changes



- Endothermic process (energy added to substance)
- Exothermic process (energy released from substance)

Energy Changes Accompanying Phase Changes

◆ Melting

- **heat of fusion** or *enthalpy of fusion*, ΔH_{fus}



◆ Vaporization

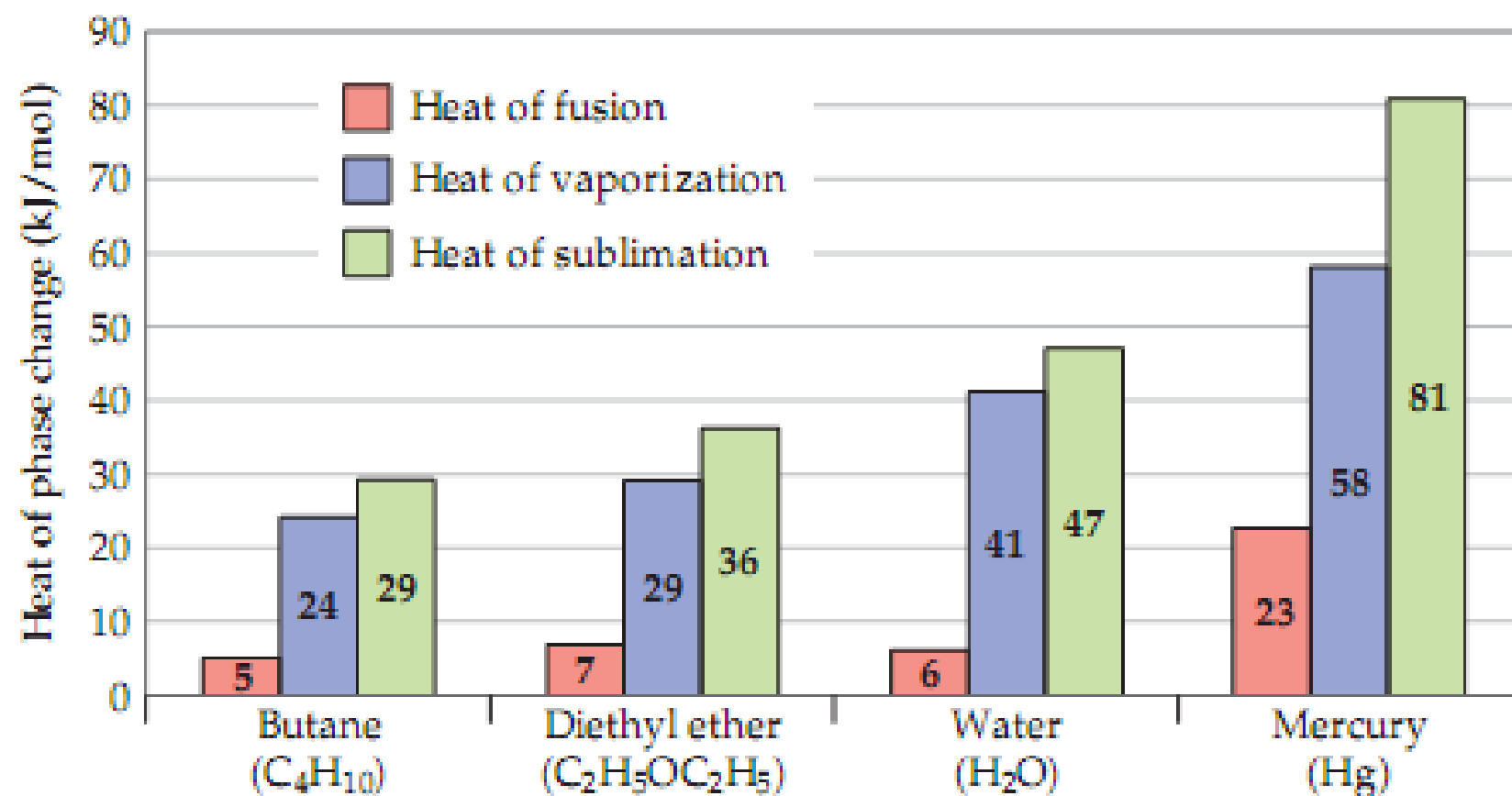
- **heat of vaporization** or *enthalpy of vaporization*, ΔH_{vap}



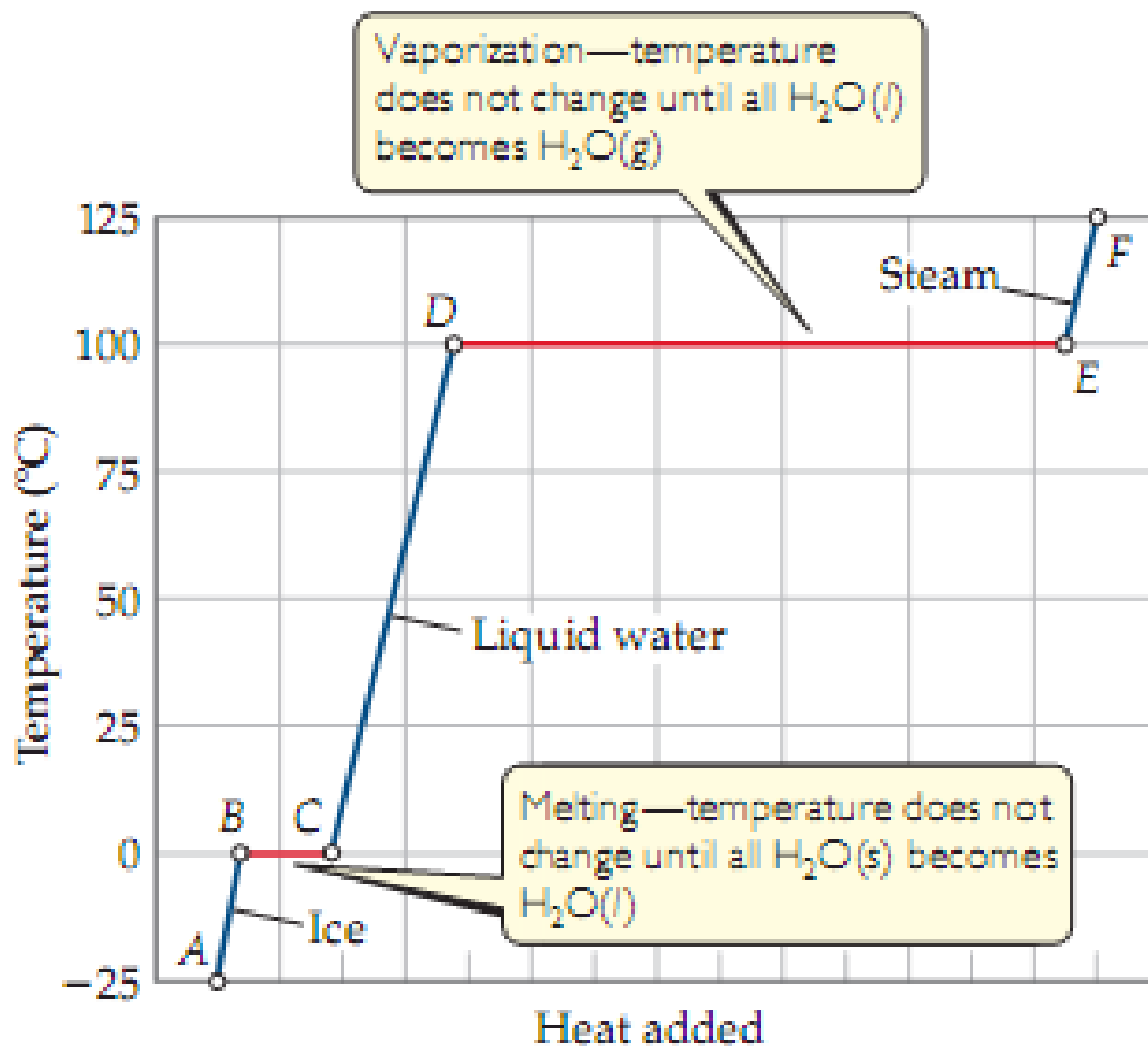
◆ Sublimation

- **heat of sublimation**, ΔH_{sub}

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$$



Heating Curves



Critical Temperature and Pressure

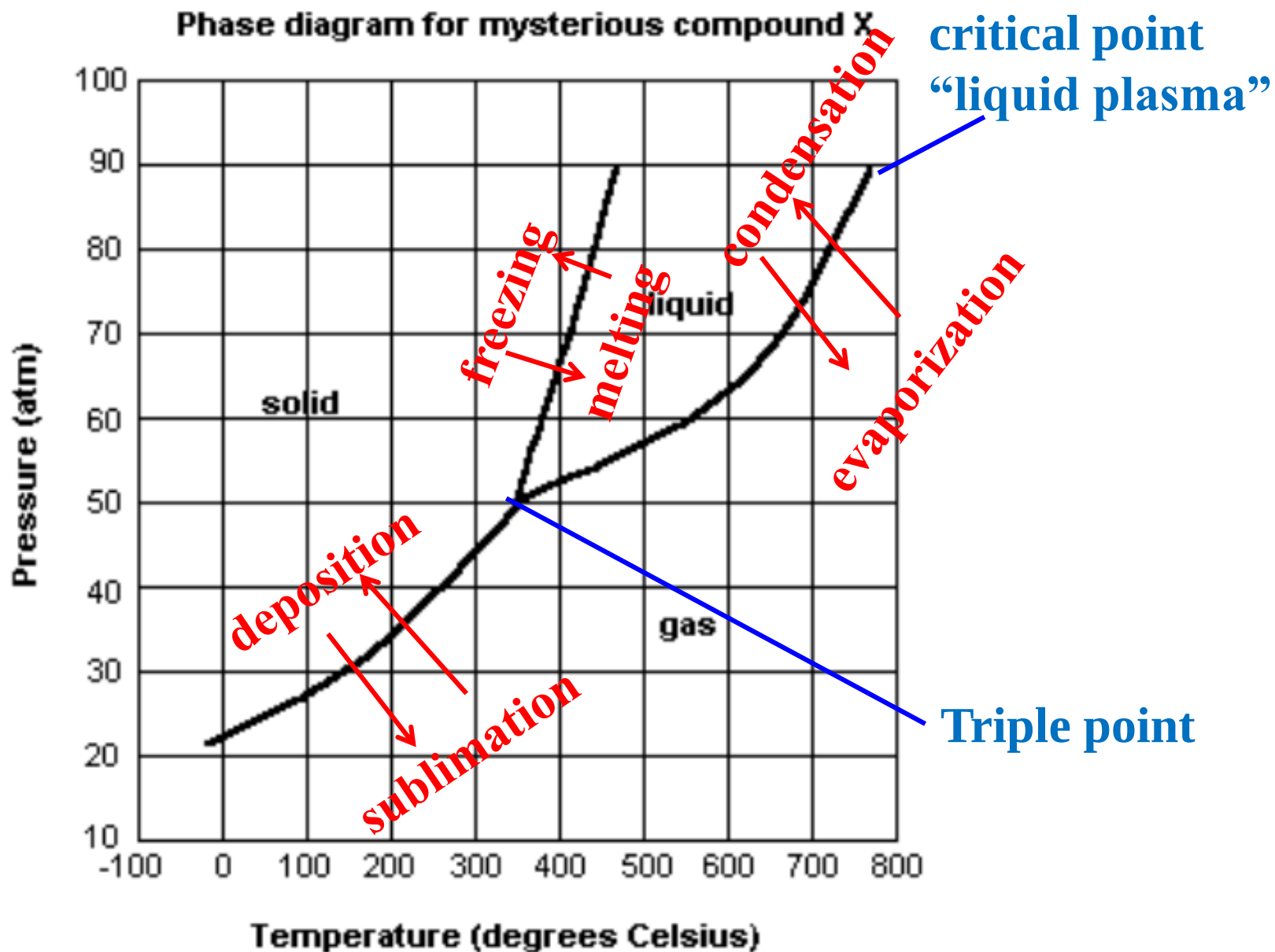
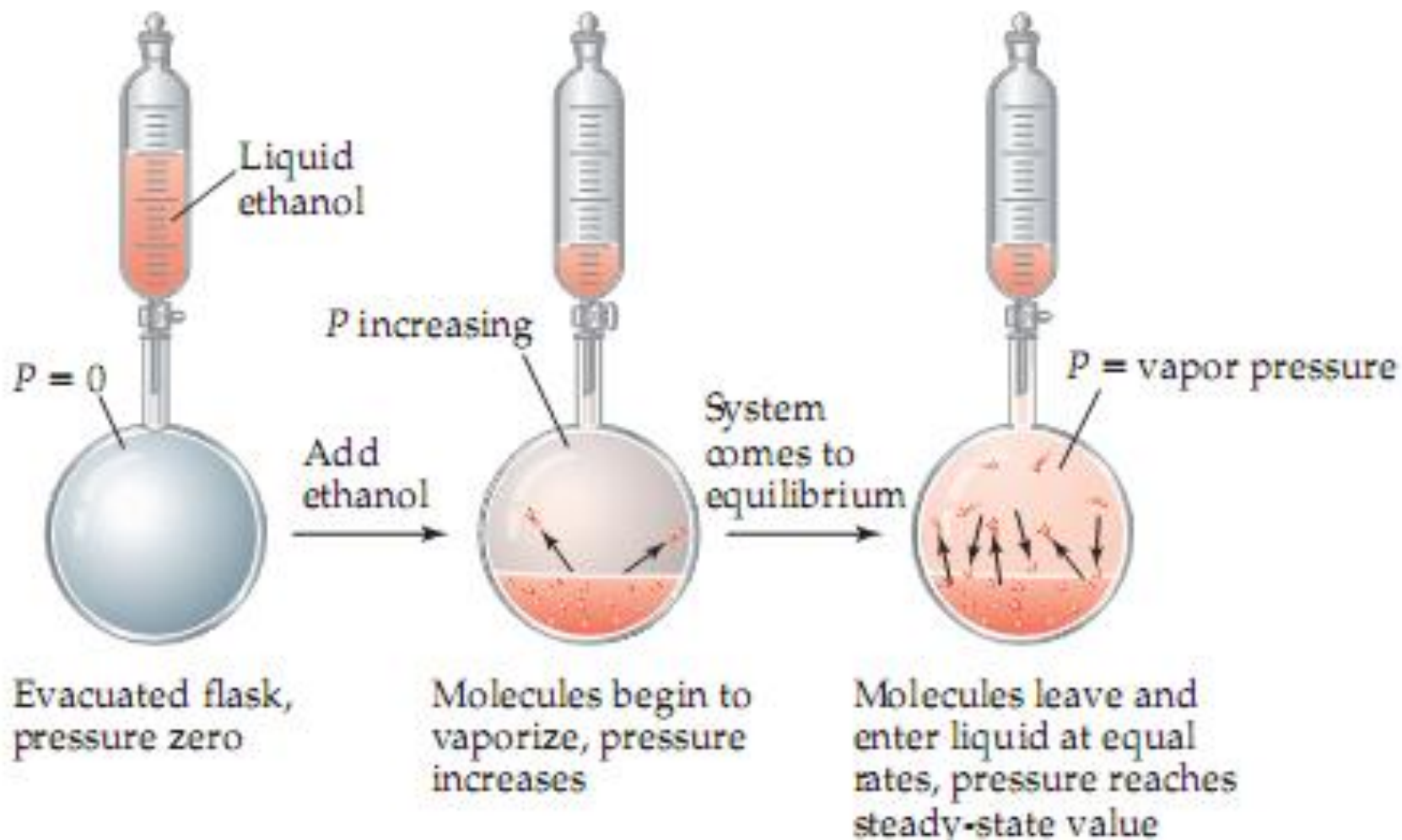


Table 11.6 Critical Temperatures and Pressures of Selected Substances

Substance	Critical Temperature (K)	Critical Pressure (atm)
Nitrogen, N ₂	126.1	33.5
Argon, Ar	150.9	48.0
Oxygen, O ₂	154.4	49.7
Methane, CH ₄	190.0	45.4
Carbon dioxide, CO ₂	304.3	73.0
Phosphine, PH ₃	324.4	64.5
Propane, CH ₃ CH ₂ CH ₃	370.0	42.0
Hydrogen sulfide, H ₂ S	373.5	88.9
Ammonia, NH ₃	405.6	111.5
Water, H ₂ O	647.6	217.7

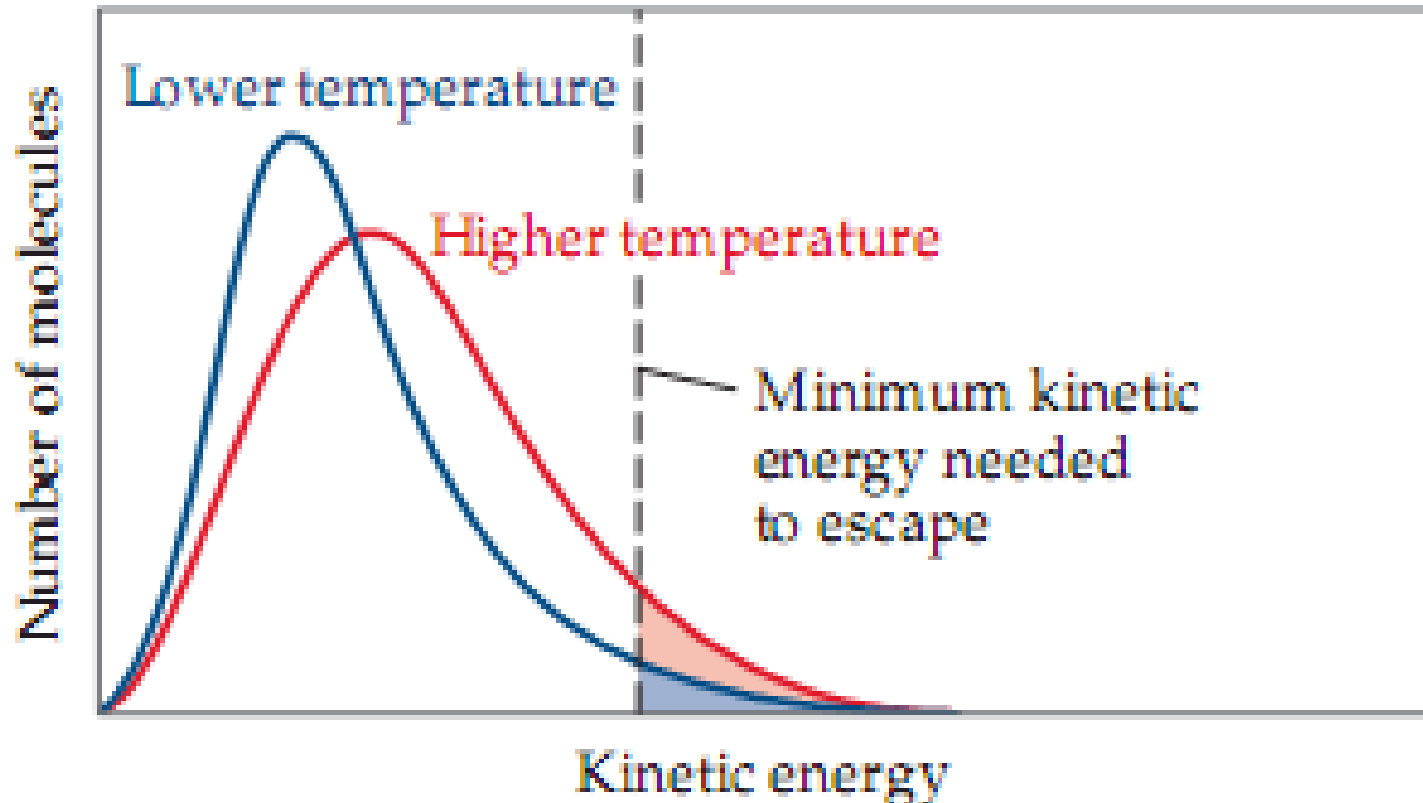
The greater the intermolecular forces, the higher the critical temperature of a substance

11-5 Vapor Pressure



The vapor pressure of a liquid is the pressure exerted by its vapor when the liquid and vapor are in dynamic equilibrium

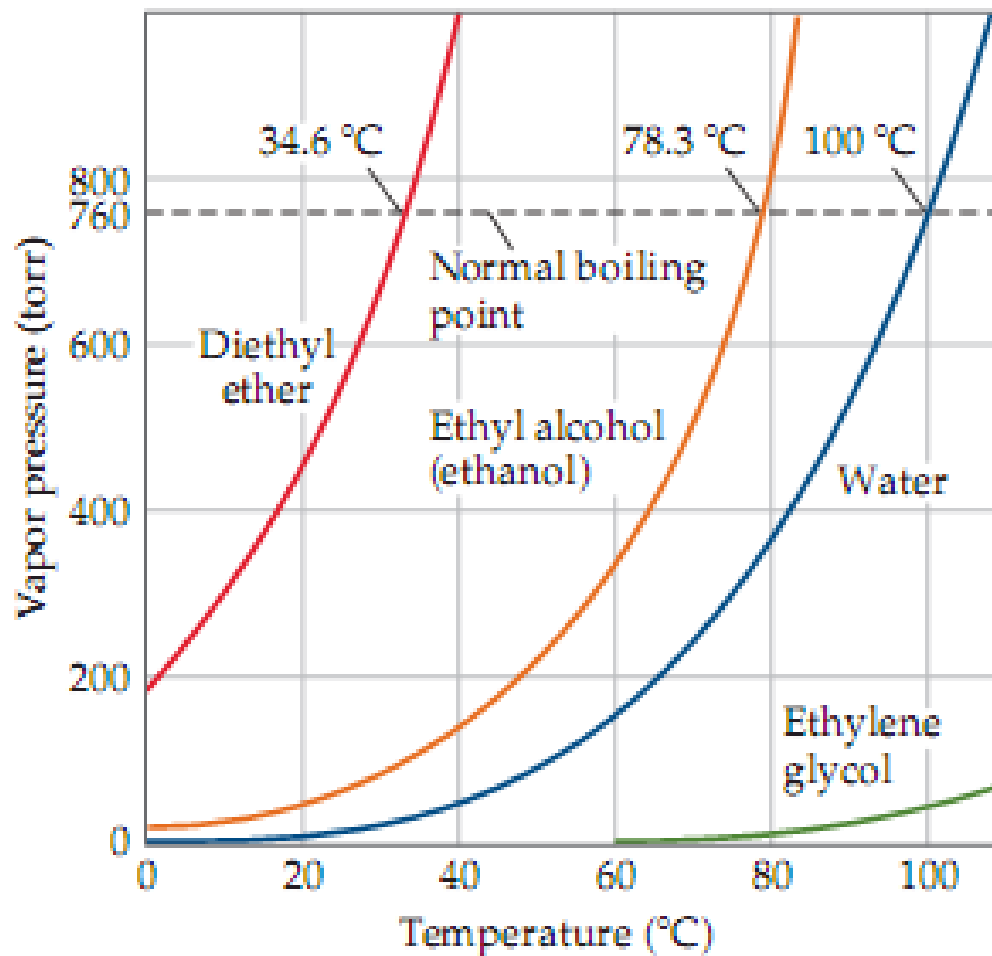
Volatility, Vapor Pressure, and Temperature



Blue area = number of molecules having enough energy to evaporate at lower temperature

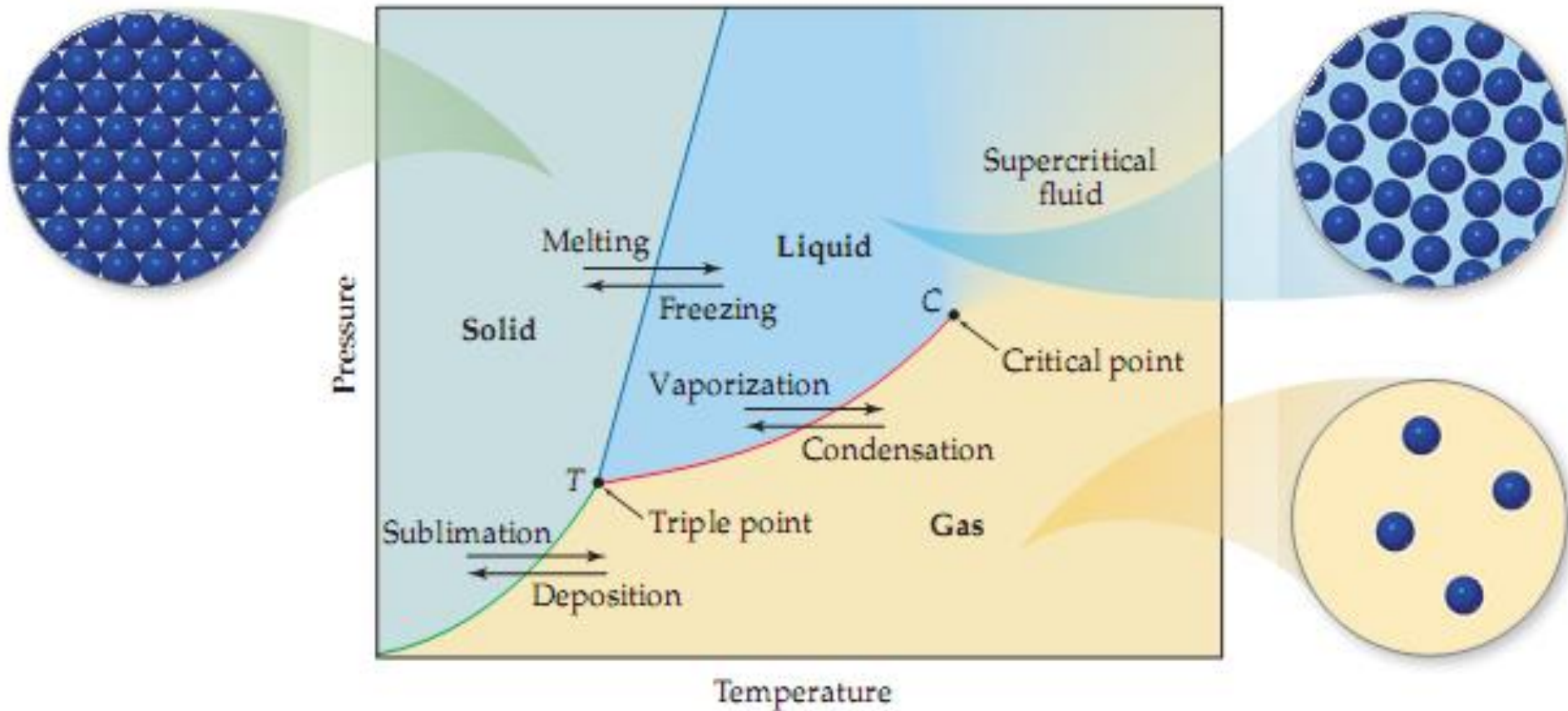
Red + blue areas = number of molecules having enough energy to evaporate at higher temperature

Vapor Pressure and Boiling Point



The boiling point of a liquid at 1 atm (760 torr) pressure is called its **normal boiling point**

11-6 Phase Diagram

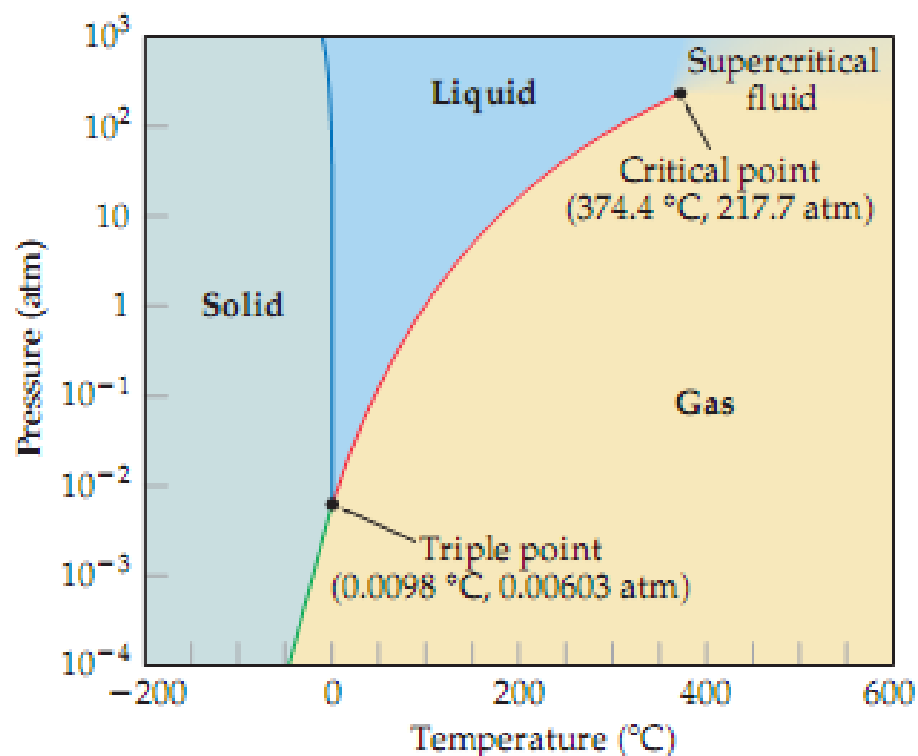


The red curve is the vapor-pressure curve

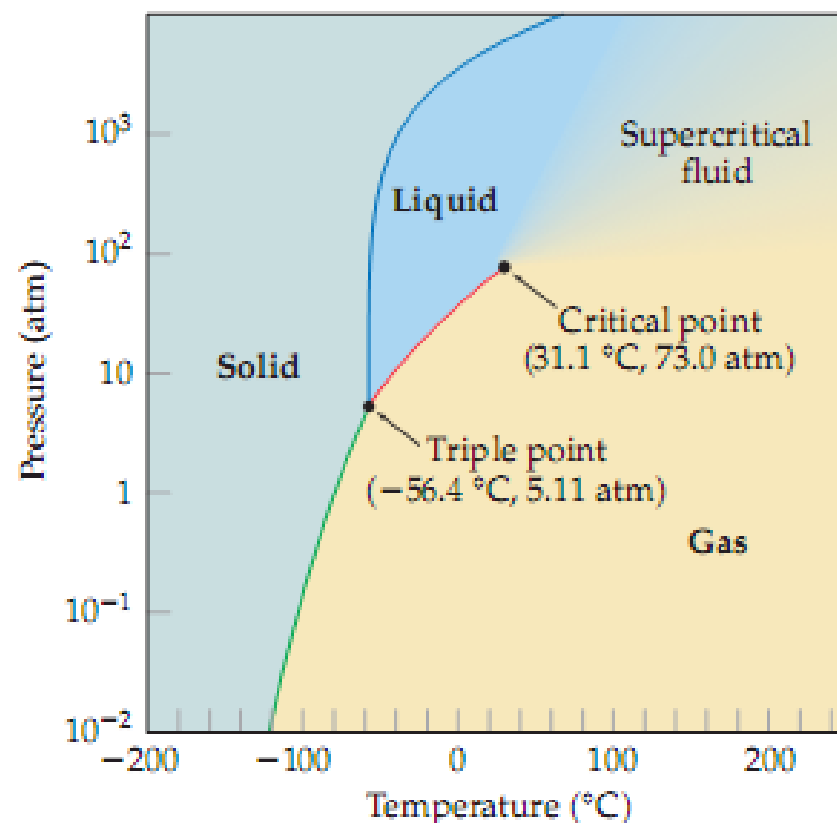
The green curve is the sublimation curve

The blue curve is the melting curve

The Phase Diagrams of H₂O and CO₂



Phase diagram of H₂O



Phase diagram of CO₂.

Homeworks

11.92

11.96

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The Central Science
Fourteenth Edition in SI Units

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